FRORE SYSTEMS

Burning Cash:

The Hidden Cost of Ignoring Thermal in Your AI Strategy

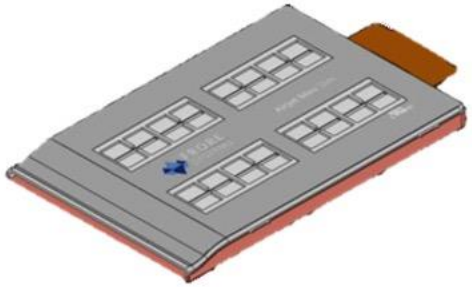
Dr. Seshu Madhavapeddy
CEO and Founder, Frore Systems
October 2026

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Frore Systems

Silicon Valley Unicorn Startup delivering thermal solutions from cloud to edge.



AirJet® solid-state active cooling chip for edge devices.



LiquidJet® 3D ultra short-loop multi-stage coldplate for AI Factories.

Industry Recognition:

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There is Only One AI Factory Metric That Matters

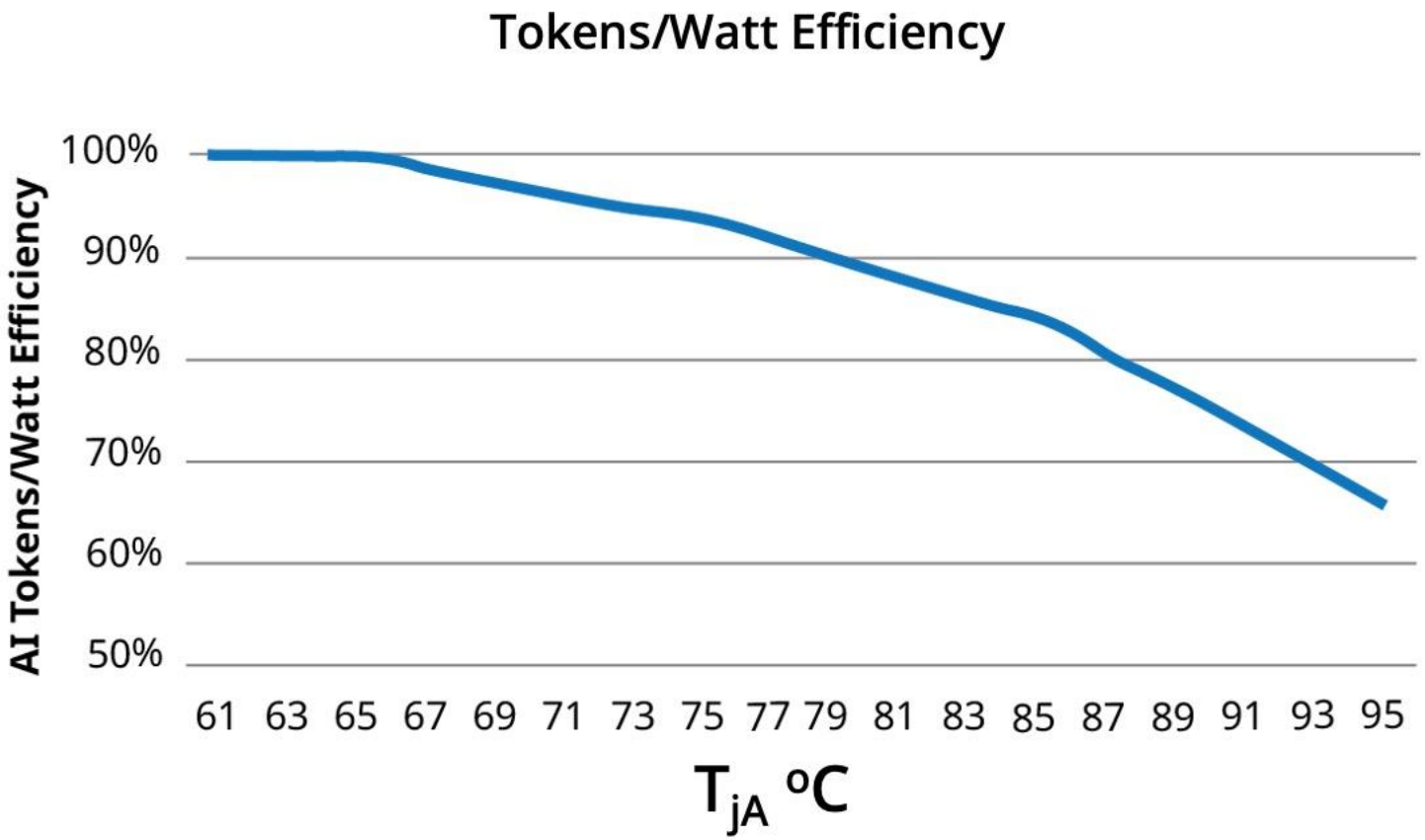
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Tokens/Watt

- Tokens are the currency of intelligence
- The world is power constrained

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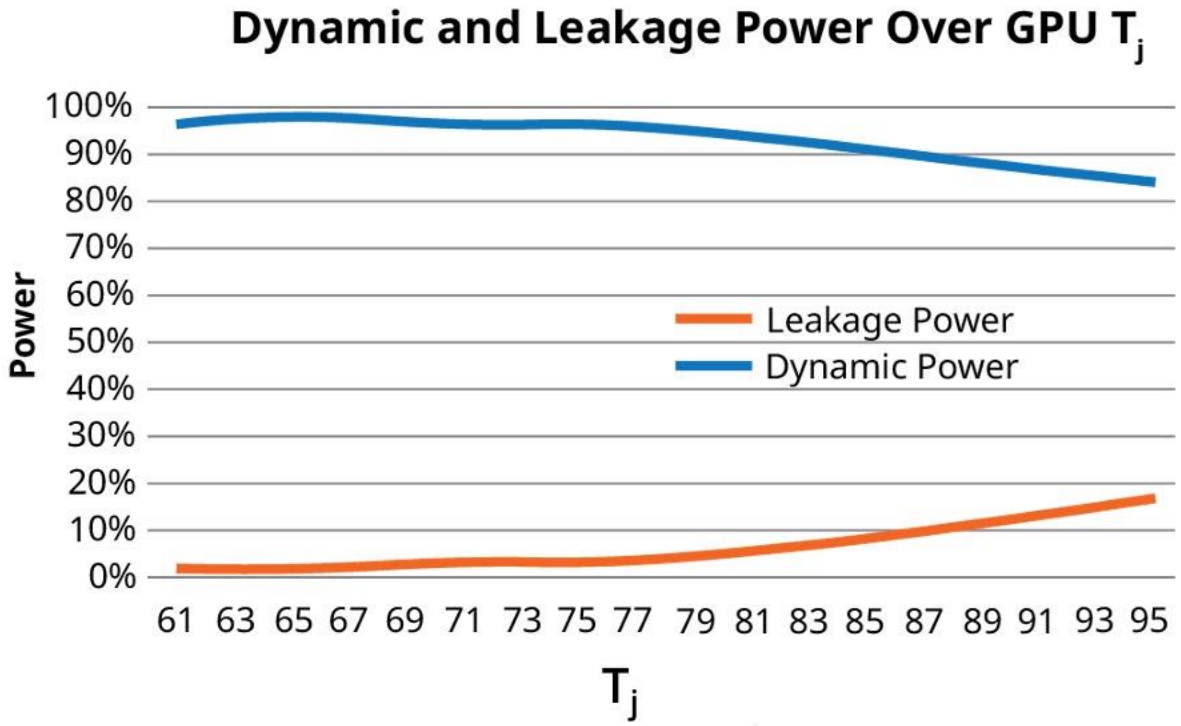
Tokens/Watt plummets with increased GPU Temperature



T_{jA} = weighted average die temperature across the entire GPU

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Semiconductor Physics



Leakage power = wasted power

Dynamic power = useful power

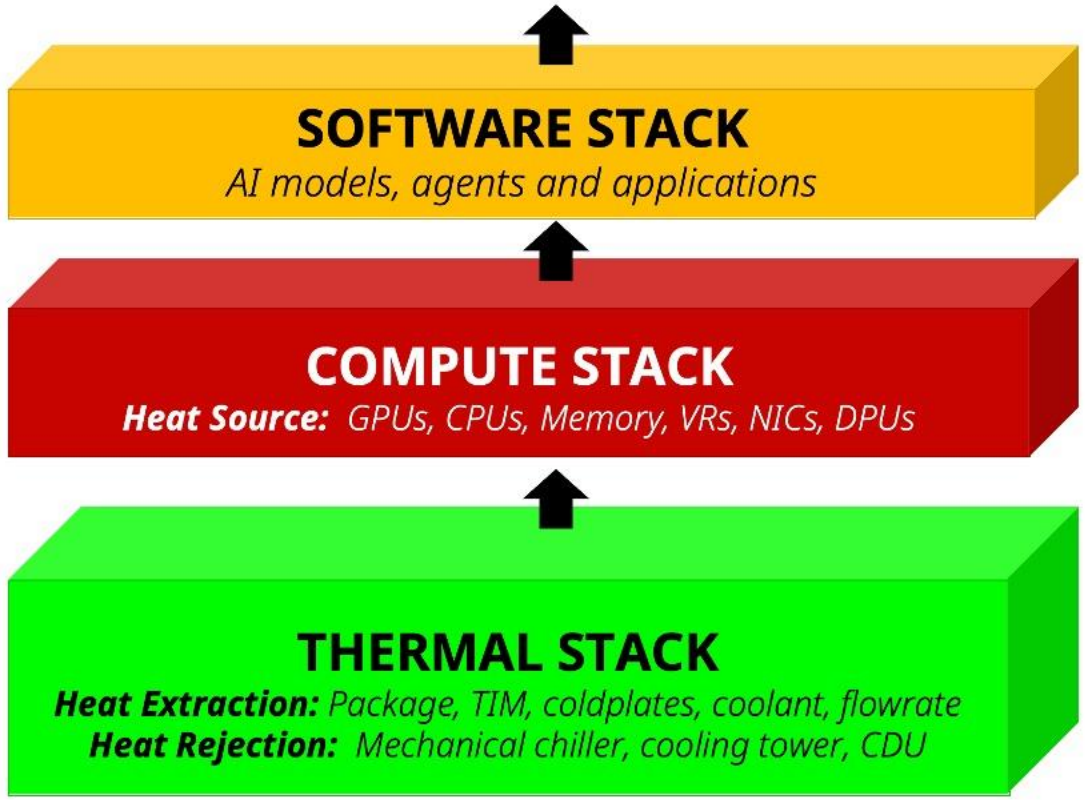
As junction temperature rises, leakage power increases, reducing available dynamic power.

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AI Factory Architecture

Cooling enables **Compute Performance** which unlocks **AI Capabilities**



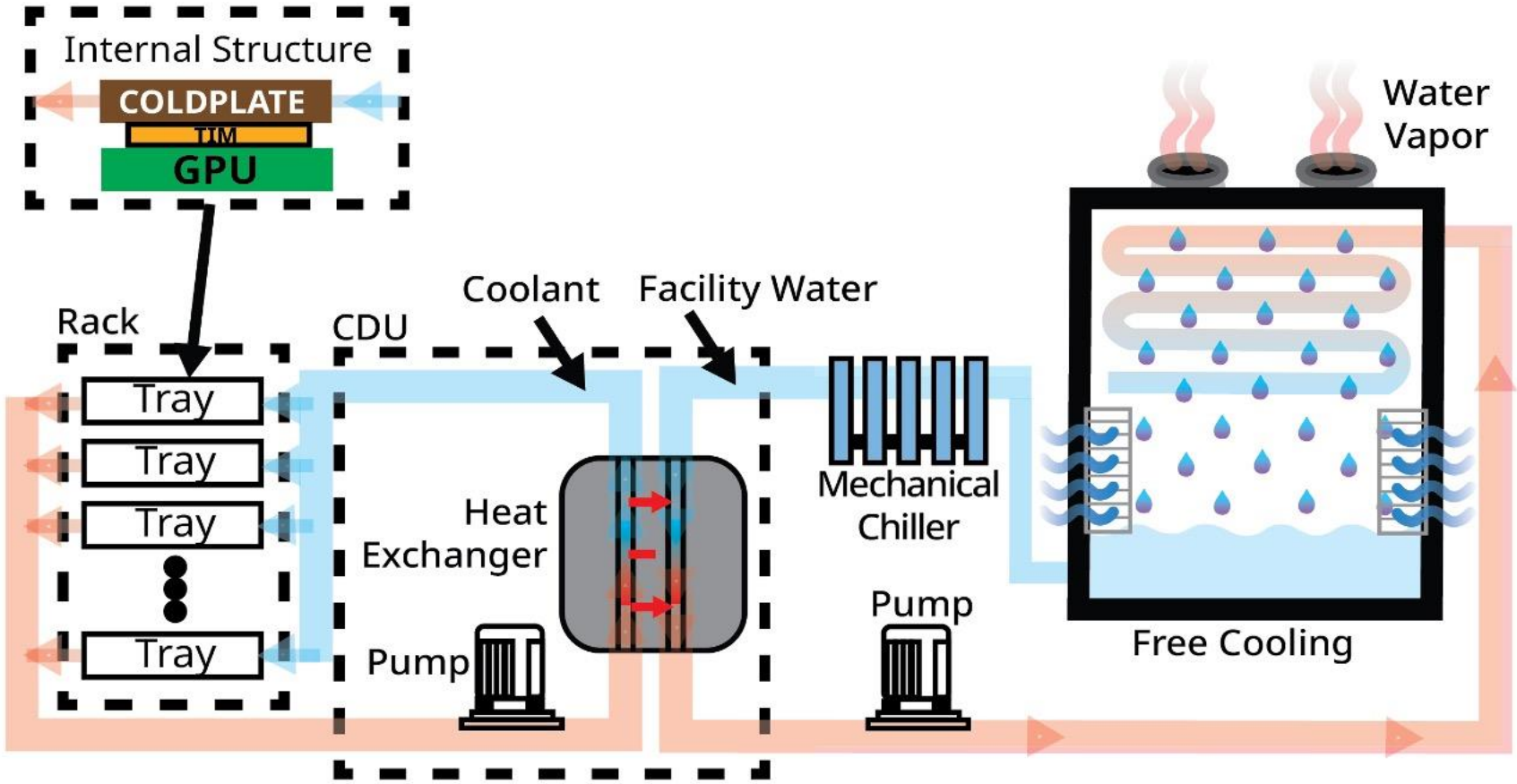
Thermal Stack is the remaining lever for AI Factory efficiency

A well-engineered Thermal Stack can increase Tokens/Watt by over 40%

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The Thermal Stack

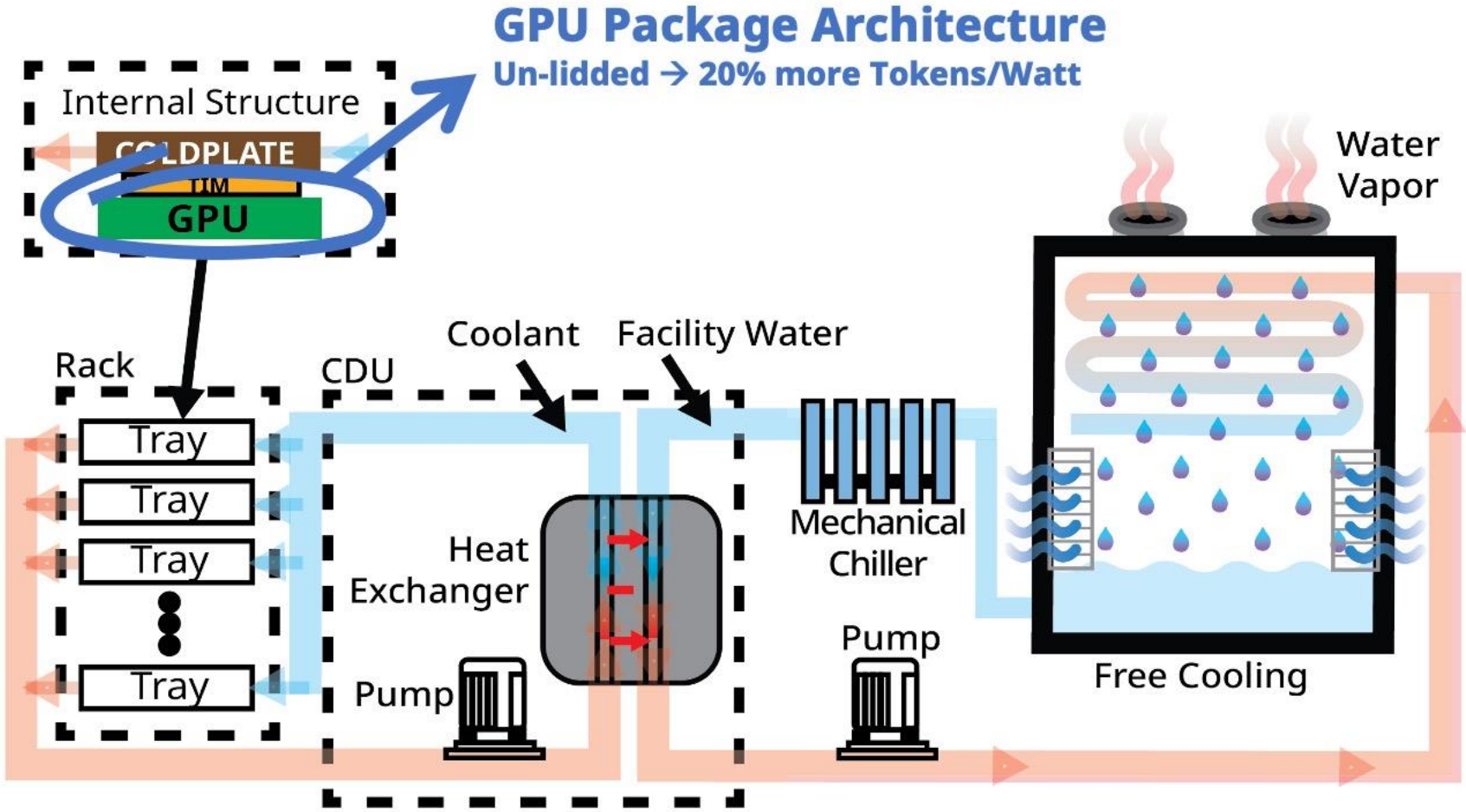
GPU → TIM → Coldplate → Coolant → Free Cooling → Mechanical Chiller



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The Thermal Stack

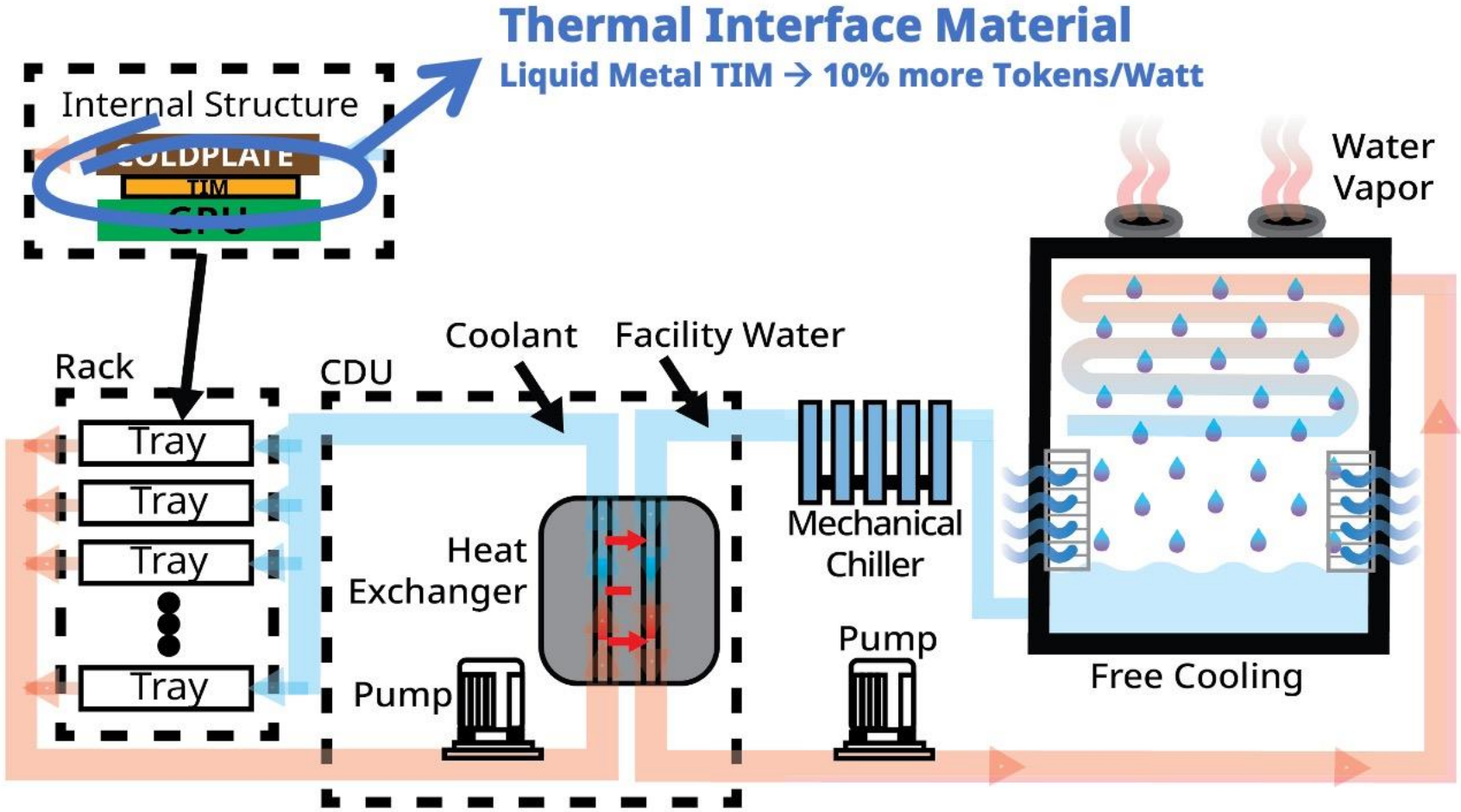
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The Thermal Stack

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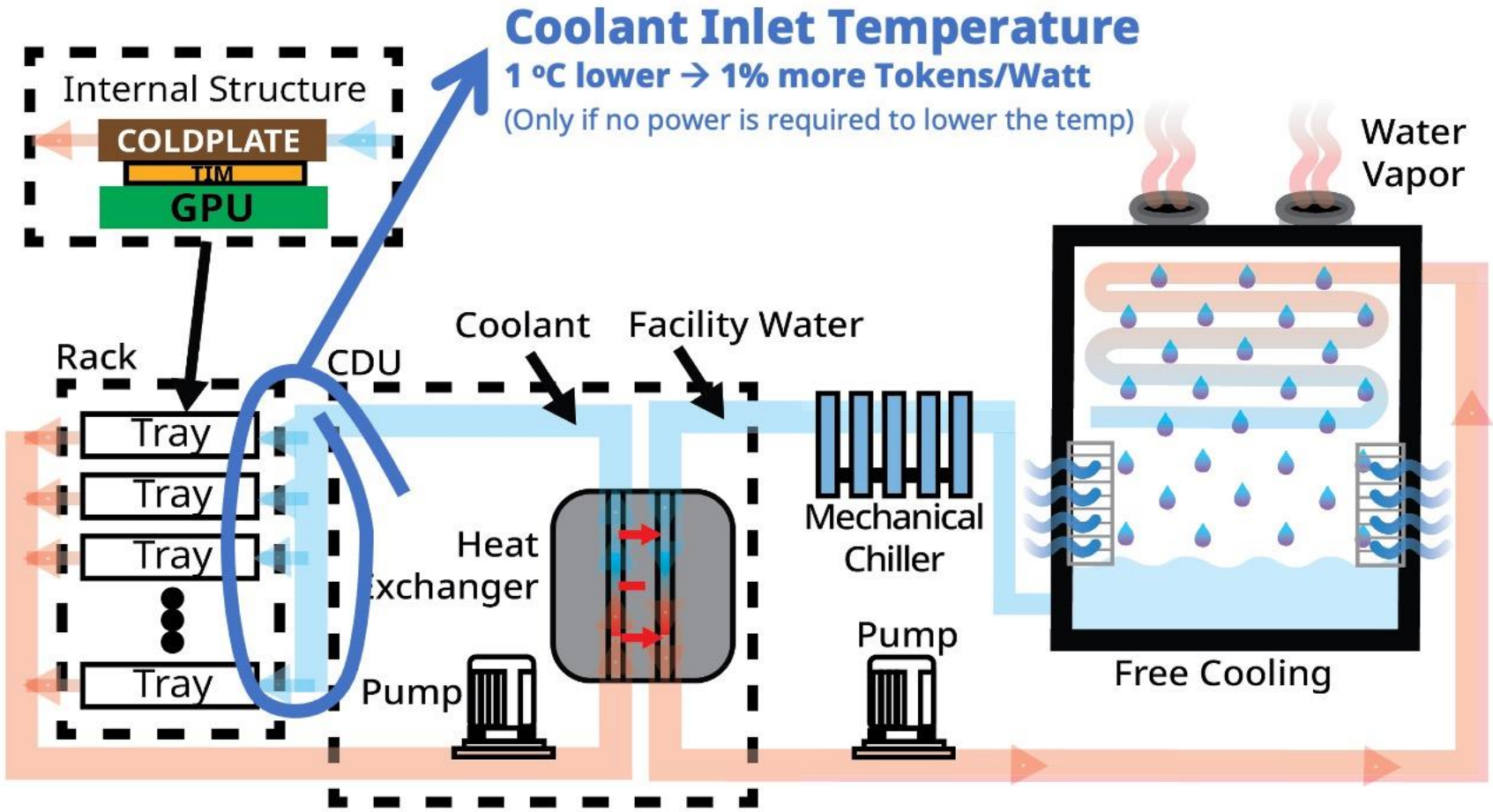


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The Thermal Stack

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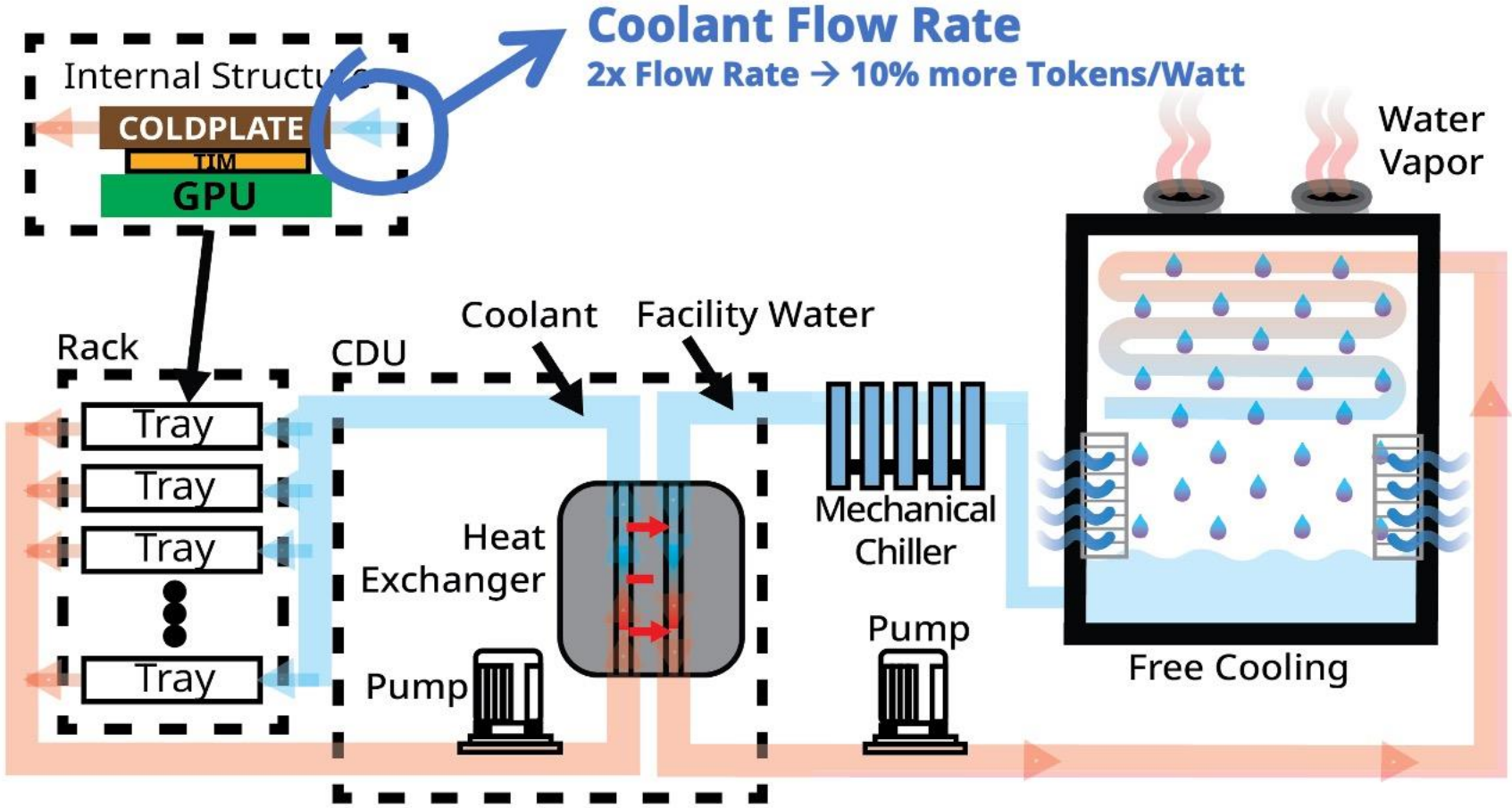


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The Thermal Stack

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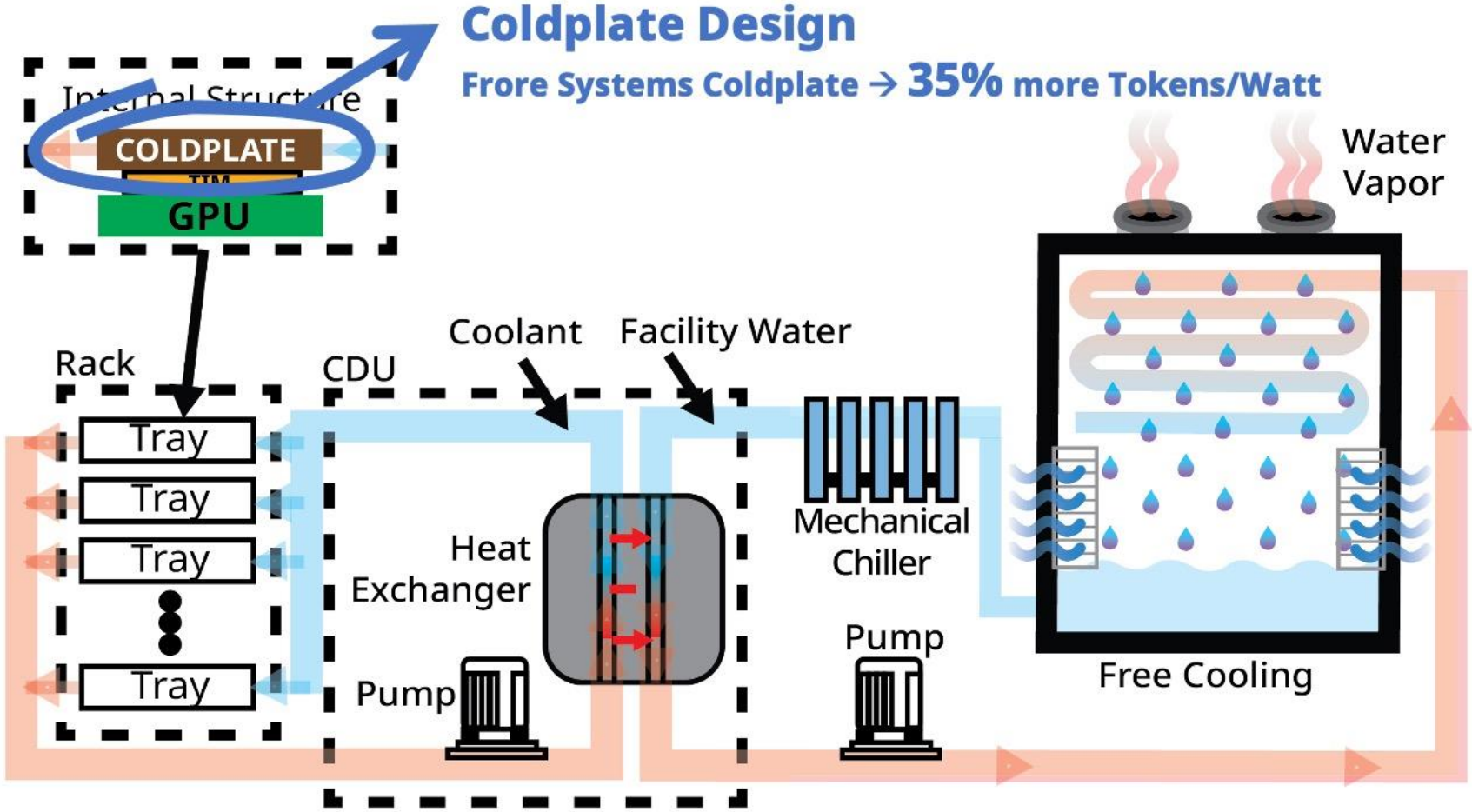


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The Thermal Stack

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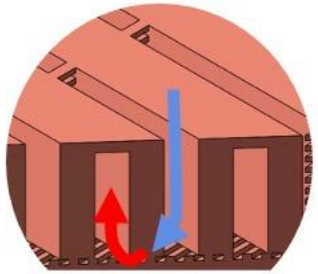
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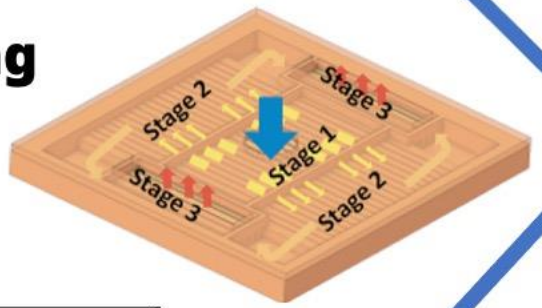
Frore Systems LiquidJet Coldplate

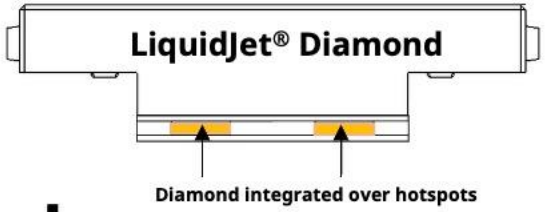


3D Ultra short loops



Multi-staging





Integrated diamond spreader

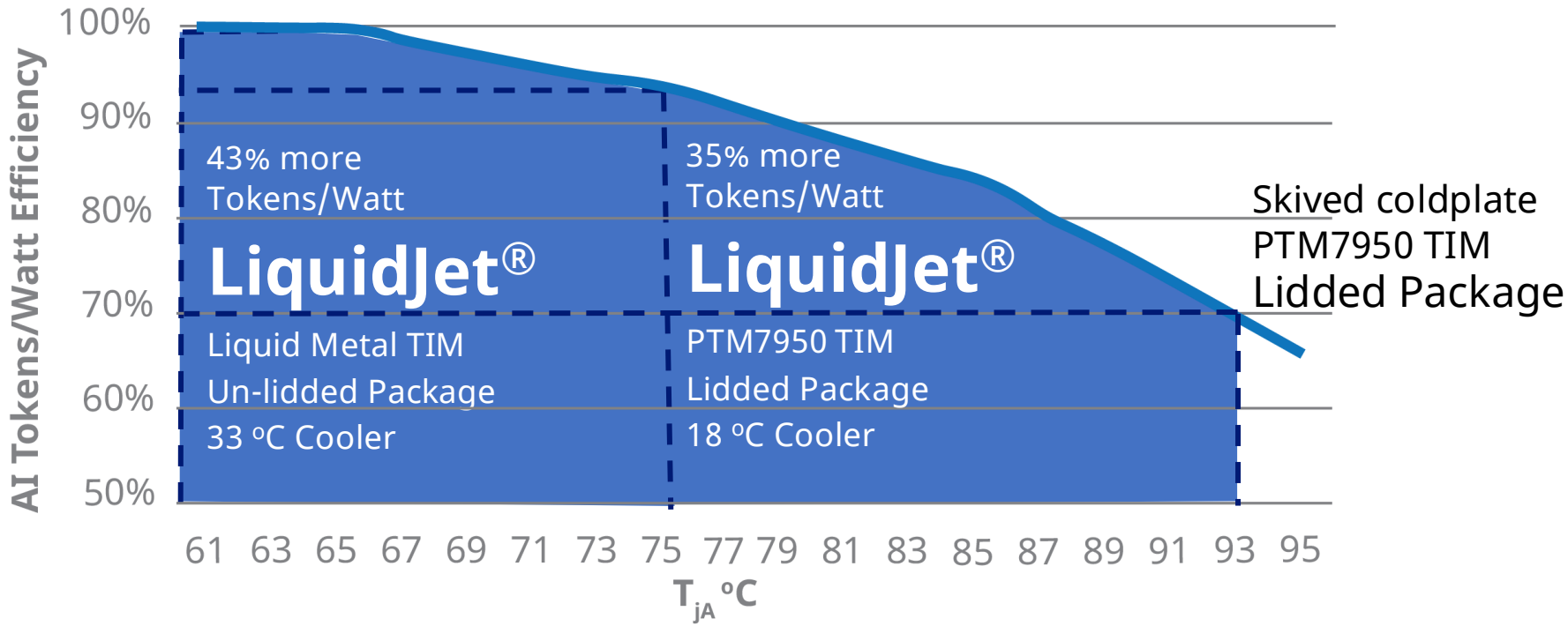
16 – 22 ºC
cooler GPUs

30 - 35%
more Tokens/Watt

Manufactured using semiconductor processes applied to copper wafers

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The Thermal Stack's Impact



A well-engineered Thermal Stack can increase Tokens/Watt by over 40%

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The AI race will be won by those who maximize Tokens/Watt by optimizing the Thermal Stack

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Thank you

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